

Sangeeta Sastra Sudharnavam: Gamakas (Musical Ornaments)

56. Gamakas

Sloka: *Tāsāmtu sthāya vāgēṣu viṣṛtiṣa vidhāsyatē || Saṃ||Ra||* Sloka: *Ētēṣāṃ miśraṇānmiśrasyāsyu rbhūrayōbhidhā :|| Saṃ||Ra||*

Sthaya = Component of a Raga; Vāga = Gamaka.

Sthāya-vāga refers to Ragas that have Gamakas as their components. There are 96 Sthaya-vagas. Sarngadeva explained that through various combinations (Sankeerna), these have evolved into infinite forms. Based on this, it can be said that Gamakas are infinite, which is also a matter of practical experience.

Gamakas are the life and soul of Indian music. They provide the unique distinction that Indian music holds among the musical traditions of the world. There are no Ragas without Gamakas. The emotional essence (Raga Bhava) and the form of a Raga are created through Gamakas. Even if two Ragas belong to the same Melakarta (parent scale) and share the same Murchana (scale), the form and nature of those Ragas change based on the variation of Gamakas on one or two notes.

Examples:

- In **Surati** and **Kedaragaula**, the notes *Sa Ni Dha Pa* share the same positions. However, when singing *Ni Dha Pa*, in **Surati** it sounds as *Nee Ni Dha Pa*, whereas in **Kedaragaula** it sounds as *Ni Dha Dha Ni Pa*.
- Ragas like **Arabhi**, **Sama**, **Kedaragaula**, **Devagandhari**, **Sankarabharanam**, **Kambhoji**, **Harikambhoji**, and **Bilahari** share similar note positions, yet they shine as distinct forms due to the subtle variations in the Gamakas used.

Gamakas are also the reason why Indian music is divided into two distinct styles. In **Hindustani (North Indian) Music**, the **Jaroo** (Glide) is of primary importance. Additionally, the Hindustani style is characterized by sustaining the notes firmly (stasis).

In contrast, **Carnatic Music** emphasizes **Kampitam** (trembling), **Oopu** (swing), and **Andolitam** (oscillation) between two notes. These types of graces (Graces/Gamakas) are unique to Indian music and are not found in other musical systems.

Although it is said that Gamakas are infinite, ancient scholars defined them as fifteen (**Panchadasa**), and later scholars categorized them into ten (**Dasha Vidha**).

57. The Fifteen (Panchadasa) Gamakas

1. **Kampita**: Trembling or vibrating a note. The oscillation should stay on the note itself without swinging too high or too low, and without touching adjacent notes (Vibrato).
2. **Lina**: Two notes merging softly and smoothly into one another. For example, in Raga Kalyani, lingering on Nishada (Ni) and merging it into Shadja (Sa). Usually, this applies from a lower note to a higher note, though the reverse is also possible.
3. **Andolita**: A prominent Gamaka in Carnatic music. It is a rhythmic oscillation between two notes.
 - Oscillating between Antara Gandhara and Panchama results in Suddha Madhyama.
 - Between Chatusruti Rishabha and Suddha Madhyama results in Sadharana Gandhara.
 - Between Chatusruti Dhaivata and Shadja results in Kaisiki Nishada.
 - This oscillation happens rhythmically and includes variations like Sukshma (subtle), Ati-sukshma (very subtle), Laghu (light), and Bruhat (large) depending on the context.
4. **Plavita**: A variation of the Andolita Gamaka.

5. **Sphurita:** Stressing or accenting the second note in a pair of identical notes (e.g., *Sa Sa, Ri Ri, Ga Ga*). This is very important and appears in almost all Ragas. Without this, the singing sounds weak.
6. **Pratyaghata (or Pratyahata):** A part of Sphurita, but used in descending order (Avarohana). For example, *Ri Sa Sa, Ni Ni Dha Dha, Pa Pa Ma Ma Ga*. It involves adding a higher note before the pair and stressing the second note.
7. **Tirupama (or Tiripemu):** Applying pressure or a "squeeze" to a note.
8. **Ahata:** Two types: (1) Rava; (2) Khandimpu.
 - **Rava:** Like *Ri Sa Sa, Sa Ni Ni*. It is essential for the "fragrance" of Carnatic music. Adding a note above a pair creates Rava.
 - **Khandimpu:** In Raga Arabhi, when singing *Sa Ni Dha* or *Ma Ga Ri*, if it is sung as *Sa Sa Dha* or *Ma Ma Ri*, the notes *Ni* and *Ga* are "cut" or skipped, creating an aesthetic effect.
9. **Vali:** Circling a note to reach the higher note. This is primarily a Veena technique.
10. **Ullasita:** Two types: (1) Ekku-jaru (Upward glide); (2) Digu-jaru (Downward glide).
 - **Ekku-jaru:** Gliding upwards (e.g., in the Saranga kriti 'Nivadanegana').
 - **Digu-jaru:** Gliding downwards (e.g., in Sahana raga, *Pa - Ri Ga Ma Pa*).
11. **Humpita:** Singing with a humming sound (like "Hum"). Is this truly a Gamaka?
12. **Kurula:** Singing notes with a "swing" (Swing). Two types: (1) Odugu; (2) Orayika.
13. **Tribhinna:** A Veena technique playing three notes on three different strings simultaneously. This is more a manner of playing than a Gamaka.
14. **Mudrita:** Singing with closed lips (making an "Ooo" sound).
15. **Namita:** Modulating or thinning the sound (Modulation).
16. **Mishrita:** A mixture of two or three of the above Gamakas.

Ancient scholars defined Gamaka as the movement, oscillation, or interaction between notes that pleases the mind of the listener (*Svarasya kampō gamakaḥ śrōtru citta mudāvahaḥ*).

58. The Ten (Dasha Vidha) Gamakas

- 1. Arohanam; 2. Avarohanam; 3. Murchana; 4. Dhalu; 5. Sphuritam; 6. Kampitam; 7. Ahatam; 8. Pratyaghatam; 9. Tripuchchham; 10. Andolitam.

Note: Arohana, Avarohanam, and Murchana are not actually Gamakas. Tripuchchha is a Veena technique. Many of these overlap with the 15 Gamakas. Modern scholars suggest that these traditional names often don't match contemporary singing practices and need re-evaluation.

59. Gamakas: Modern Logical Perspective

In modern practice, Carnatic music is characterized by the prevalence of **Andolita** (oscillation). While Hindustani music relies heavily on the **Jaru** (Glide), the rhythmic swing between two notes is the hallmark of the Carnatic style.

Technical Comparisons: Hindustani vs. Carnatic

The following examples illustrate how the same notes are treated differently in the two systems:

Note/Phrase	Hindustani Treatment	Carnatic Treatment
Nishada (Ni)	<i>Sa . Ni , Sa , Ni</i>	<i>Sa Da Da Sa Ni Sa</i>
Sadharana Ga	<i>Ma . x . Ma , Ga</i>	<i>Ma , Ri Ri Ma G Ma</i>
Suddha Madhyama	<i>Pa , Ma</i>	<i>Pa , Ma Pa G Ma Pa</i>

Suddha Rishabha	<i>Ri Sa , Ri (Komal)</i>	<i>Sa , Ri Sa , Ri (Suddha)</i>
Suddha Dhaivata	<i>Ni Dha Ni Dha (Komal)</i>	<i>Pa Dha Pa Dha (Suddha)</i>

Traditional Practice of "Non-Flat" Notes

In the Carnatic tradition, it is considered contrary to established custom to hold certain notes "flat" or without ornamentation. These notes must always be supported by adjacent "anchor" notes:

- Suddha Rishabha (Ri1): Anchored to Shadja (*Sa Ri Sa Ri*).
- Sadharana Gandhara (Ga2): Often sung as *Ri Ma Ga Ma*.
- Suddha Dhaivata (Dha1): Anchored to Panchama (*Pa Dha Pa Dha*).
- Kaisiki Nishada (Ni2): Often sung as *Dha Sa Ni Sa*.
- Prati Madhyama (Ma2): Anchored to Panchama (*Pa Ma Pa Ma*).

Specific Gamaka Patterns by Note

The text provides a practical guide for how specific swaras are typically ornamented:

- Suddha Rishabha (Shu.Ri): *Sa Ri Sa Ri* or *Ri Sa Ri Sa*.
- Chatusruti Rishabha (Cha.Ri): *Sa Ga Ri* or *Sa Ma Ri* or *Ga Ri Ga Ri*.
- Suddha Dhaivata (Shu.Dha): *Pa Dha Pa Dha* or *Dha Pa Dha Pa*.
- Chatusruti Dhaivata (Cha.Dha): *Pa Ni Dha* or *Pa Sa Dha* or *Ni Dha Ni Dha*.
- Antara Gandhara (Ant.Ga): *Ma Ga Ma Ga* or *Ga Ma Ga Ma*.
- Prati Madhyama (Pra.Ma): *Pa Ma Pa Ma* or *Ma Pa Ma Pa*.

Categories of Oscillation

Andolita is further divided based on the width of the swing:

- Bruhadandolita (Big Oscillation): A wide swing, e.g., *Ma = Ga Ma Ga Sa Ma Pa*.
- Svalpandolita (Small Oscillation): A narrow, subtle swing, e.g., *Ma = Ga Ma Ga Ma*.

Raga Formation and Components

Gamakas do not exist in isolation. They combine with other elements to form a Raga:

1. **Gamakas:** The primary ornaments (oscillations, glides, etc.).
2. **Alankaras:** Defined groups or sequences of notes.
3. **Varnas:** The melodic movements (Arohi, Avarohi, Sthayi, Sanchari).
4. **Sangatis:** Melodic variations formed by combining the above with subtle microtones (*Sruti*).

Case Study: The "Ma Ga Ri Sa" Phrase

The phrase *Ma Ga Ri Sa* appears in many Ragas, but its emotional essence changes entirely based on the Gamakas applied:

Raga	Gamaka Pattern
Devagandhari	<i>Ma Ga Ri Ree Sa - Sa Ri Ga - Sa Ri Sa</i>
Sama	<i>Maa Ma Ga Ri Sa Ri</i>
Surati	<i>Ma Ga Ga Ri Sa Ri Ga Ri Ree</i>

Kedaragaula

Maa Ga Pa Maa Ma Ga Ri

Yadukula Kambhoji

Ma Ga Gaa Ma Ree Ree Reesa

Sahana

Ree Ree : Ga Ma Pa Ma Maa Ga : Ga Ma Ree